

MENTAL HEALTH PREDICTION USING MACHINE LEARNING FOR HEALTHCARE WORKERS

An Explainable, Fair and Calibrated Multi-Source Framework for Stress, Anxiety and Depression Risk Prediction

Optional alternate title: Explainable, Fair and Calibrated Multi-Source Mental Health Risk Prediction for Healthcare Workers — An Evidence-Grounded Unified Framework Using Machine Learning and Generative AI

ABSTRACT

Mental health is a significant concern among healthcare workers due to demanding workloads, long working hours, high patient responsibility, irregular shifts, emotionally challenging situations and continuous exposure to stressful clinical environments. Prolonged occupational stress can contribute to burnout and may also be associated with anxiety and depression. Therefore, identifying elevated mental-health risk among healthcare workers at an early stage can support awareness and preventive intervention.

This project, Mental Health Prediction Using Machine Learning for Healthcare Workers, extends an existing machine-learning-based stress and burnout prediction system developed during the previous semester. The previous system primarily focused on stress and burnout risk using physiological, environmental and occupational information. The proposed project expands the scope toward broader mental-health risk prediction by incorporating the DASS-42 dataset, which provides psychological indicators associated with depression, anxiety and stress, and by incorporating an additional healthcare workforce mental-health dataset that provides workforce-level context.

The proposed framework utilizes four complementary datasets: the Stress-Lysis dataset, the healthcare workplace survey dataset, the DASS-42 dataset and the Healthcare Workforce Mental Health dataset. The Stress-Lysis dataset contributes physical and environmental information such as temperature, humidity and activity level. The workplace survey contributes occupational information including weekly working hours, patients per shift, department and stress score. DASS-42 contributes psychological information through its depression, anxiety and stress questionnaire responses. The workforce dataset contributes workplace stressor and workforce mental-health outcome information for organization-level analysis and robustness experiments.

Because these datasets originate from different sources and do not represent the same individuals, they will not be directly concatenated row-by-row. Instead, the project will develop a unified multi-source feature representation in which physical, environmental, occupational, psychological and workforce information is standardized and transformed into complementary features for the mental-health prediction framework.

Five machine-learning algorithms will be implemented and compared: Decision Tree, Random Forest, XGBoost, LightGBM and Multilayer Perceptron (MLP). Their performance will be evaluated using accuracy, precision, recall, F1-score, confusion matrices and other appropriate classification metrics.

The evaluation is further extended with probability calibration (reliability diagrams, Brier score, Expected Calibration Error), fairness assessment using Fairlearn, class-imbalance experiments using class weighting and SMOTE, and ablation studies that quantify the contribution of each dataset, feature group and methodological component.

To improve interpretability, SHAP-based Explainable AI will be used to identify the factors contributing to individual and overall predictions, and DiCE will be used to generate counterfactual "what-if" explanations. A Generative AI component, grounded through Retrieval-Augmented Generation (RAG), will then convert model predictions and feature explanations into understandable mental-health risk summaries for healthcare workers, together with an organization-oriented workforce report.

The existing Python, Jupyter Notebook and Streamlit architecture will be extended rather than replaced with a separate frontend, backend or database system, with Plotly added for interactive analytics and MLflow as an optional experiment-tracking layer. The final system will serve as an experimental and educational framework for mental-health risk prediction among healthcare workers and will not be intended as a clinical diagnostic system.

CHAPTER 1 — INTRODUCTION

1.1 Background

Healthcare workers operate in environments that can involve high workloads, long working hours, emergency situations, irregular schedules, patient responsibility and emotionally demanding interactions. These conditions can contribute to elevated occupational stress and may affect overall psychological well-being.

Mental health among healthcare workers is therefore an important area for research. Stress, anxiety and depression are interconnected but distinct dimensions of psychological well-being. A system that focuses only on stress or burnout may not capture the broader mental-health profile of a healthcare worker.

Machine learning provides an opportunity to identify patterns in structured data and estimate the likelihood of different mental-health risk categories. However, a prediction alone is not sufficient for a responsible system. The predicted probability should be reliable, the errors should be examined across groups, and the reasoning behind the prediction should be explainable.

The proposed project focuses specifically on healthcare workers and extends the existing stress and burnout prediction system into a broader, responsible mental-health prediction framework.

The system will investigate three major dimensions:

Stress

+

Anxiety

+

Depression

using information from physical, environmental, occupational, psychological and workforce sources.

1.2 Existing Project

The previous semester's project, *Explainable Generative AI for Stress and Burnout Risk Prediction in Healthcare Workers*, focused on identifying stress and burnout-related patterns among healthcare workers.

The existing project utilized two major datasets.

Stress-Lysis Dataset

The Stress-Lysis dataset consists of approximately 2,001 records containing:

Ambient humidity

Ambient temperature

Number of steps

Stress level

The stress variable is represented using three levels:

Low

Medium

High

Healthcare Workplace Survey

The workplace survey contains approximately 150 records across ten hospital departments:

OPD, IPD, ICU, ER, OBG, PED, MED, SRG, DER, RHEUN

The dataset contains variables such as:

Weekly work hours

Patients per shift

Department ID

Stress score

The previous system used:

Random Forest

XGBoost

MLP

SHAP

Generative AI

Streamlit

1.3 Proposed Expansion

The proposed project expands the existing system from:

Stress and Burnout Risk Prediction

to:

Explainable, Fair and Calibrated Mental Health Risk Prediction among Healthcare Workers

The major additions are:

The DASS-42 dataset (psychological dimension).

A fourth Healthcare Workforce Mental Health dataset (workforce dimension).

A Responsible AI layer — calibration and fairness.

Counterfactual explanations using DiCE alongside SHAP.

Ablation studies to provide experimental evidence for each component.

RAG-grounded Generative AI reporting.

Plotly-based interactive analytics and an optional MLflow tracking layer.

The final framework will consider:

Physical / Environmental Factors

+

Occupational Factors

+

Psychological Factors

+

Workforce Factors

↓

Mental Health Risk Prediction



Stress + Anxiety + Depression



Calibration + Fairness + Explainability



Evidence-Grounded Report

This provides a clear continuation of the previous project while significantly increasing its scope and methodological depth.

1.4 Motivation

Healthcare workers frequently work under demanding conditions. Workload, working hours, patient volume and environmental factors may influence stress levels, while psychological factors can provide additional information regarding anxiety and depression.

The previous project primarily concentrated on stress and burnout. However, mental health is broader than these two outcomes.

Therefore, the proposed project introduces DASS-42 to provide psychological information associated with depression, anxiety and stress, and a workforce dataset to provide organization-level context such as workplace stressors and workforce mental-health outcomes.

A second motivation is responsibility. A mental-health risk system that outputs a probability should be evaluated for whether that probability is meaningful, and whether its errors are distributed unevenly across groups. The project therefore aims to investigate not only *how accurately* the models predict, but also *how reliably*, *how fairly* and *how understandably*.

1.5 Problem Statement

Healthcare workers may experience substantial occupational and psychological stress due to long working hours, high patient workloads, demanding clinical environments and emotionally challenging situations.

Existing systems may focus on individual factors or a single mental-health outcome, and often report only predictive accuracy without examining probability reliability, group-level disparities or the contribution of individual components.

The revised problem is therefore to design and experimentally evaluate a unified, explainable and responsible machine-learning framework that can learn from complementary physical/environmental, occupational, psychological and workforce information, while explicitly accounting for data-source differences, class imbalance, calibration, fairness and uncertainty.

The system will also investigate whether model explanations can be converted into understandable, evidence-grounded reports without allowing the Generative AI layer to independently diagnose or determine risk.

1.6 Aim

To develop and evaluate an explainable, fair, calibrated and evidence-grounded multi-source machine-learning framework for predicting stress, anxiety and depression risk among healthcare workers using physical, environmental, occupational, psychological and workforce information.

1.7 Objectives

The major objectives are:

Extend the existing healthcare-worker stress and burnout prediction system.

Incorporate the DASS-42 dataset into the existing project.

Incorporate a fourth Healthcare Workforce Mental Health dataset for workforce-level analysis and robustness experiments.

Utilize four complementary datasets:

Stress-Lysis

Healthcare Workplace Survey

DASS-42

Healthcare Workforce Mental Health dataset

Develop a unified multi-source feature representation without assuming row-wise identity across independent datasets.

Incorporate physical, environmental, occupational, psychological and workforce factors.

Develop and compare five machine-learning models: Decision Tree, Random Forest, XGBoost, LightGBM, MLP.

Predict three major mental-health dimensions: Stress, Anxiety, Depression.

Evaluate model performance using standard classification metrics.

Handle class imbalance using SMOTE, class weighting or other justified methods, applied only to training data and only when supported by exploratory analysis.

Measure probability reliability using calibration curves, Brier score and Expected Calibration Error (ECE), and investigate post-hoc calibration.

Assess group-level fairness using appropriate Fairlearn metrics and investigate mitigation methods.

Apply SHAP for global and local explainable predictions.

Apply DiCE for constrained counterfactual explanations.

Perform ablation studies to quantify the contribution of datasets, feature groups and responsible-AI components.

Use RAG to ground Generative AI reports in retrieved evidence and supplied model explanations.

Produce both individual and organization-oriented report views.

Extend the existing Streamlit application with Plotly interactive analytics.

Optionally track experiments using MLflow.

Analyze the limitations and applicability of machine-learning-based mental-health risk prediction for healthcare workers, including synthetic-data caveats.

1.8 Research Questions

RQ1 — Can physical/environmental, occupational, psychological and workforce mental-health information be incorporated into a unified machine-learning framework for mental-health risk prediction among healthcare workers without assuming row-wise identity across datasets?

RQ2 — How do Decision Tree, Random Forest, XGBoost, LightGBM and MLP compare across the stress, anxiety and depression prediction tasks?

RQ3 — How much do different datasets and feature groups contribute to predictive performance?

RQ4 — How does class balancing (class weighting, SMOTE) affect recall, F1, calibration and fairness?

RQ5 — Are the model probabilities well calibrated, and can post-hoc calibration improve probability reliability?

RQ6 — Do prediction errors or performance metrics differ across available demographic/contextual groups?

RQ7 — Can fairness mitigation reduce measured disparities without unacceptable degradation in predictive performance or calibration?

RQ8 — Can SHAP and DiCE provide complementary explanations for model predictions?

RQ9 — Can RAG-grounded Generative AI produce more evidence-grounded and transparent reports than an ungrounded generation approach?

RQ10 — Can the final framework support both individual-level interpretation and organization-level workforce analysis without presenting model outputs as clinical diagnoses?

CHAPTER 2 — DATASETS USED

2.1 Overview

The proposed system uses four complementary datasets.

Dataset	Main Information	Role
Stress-Lysis	Temperature, humidity, steps, stress	Physical / environmental factors
Healthcare Workplace Survey	Work hours, patients/shift, department, stress	Occupational factors
DASS-42	Psychological questionnaire responses	Depression, anxiety and stress
Healthcare Workforce Mental Health	Workplace stressors and workforce mental-health outcomes	Workforce-level analysis, additional features/targets, robustness experiments

The four datasets are not treated as if they contain the same individuals. Instead, they provide complementary information that contributes to the overall unified framework.

2.2 Stress-Lysis Dataset

The Stress-Lysis dataset consists of approximately 2,001 records.

Variables

Ambient Humidity

Ambient Temperature

Number of Steps

Stress Level

Stress Classes

0 → Low

1 → Medium

2 → High

Contribution

The dataset provides the project's physical and environmental component.

Potential derived variables include:

Activity level

Environmental stress score

Temperature-related indicator

Humidity-related indicator

Physical activity intensity

2.3 Healthcare Workplace Survey Dataset

The workplace survey contains approximately 150 records collected across ten hospital departments.

Departments

OPD, IPD, ICU, ER, OBG, PED, MED, SRG, DER, RHEUN

Variables

Weekly work hours

Patients per shift

Department ID

Stress score

Contribution

This dataset provides the healthcare occupational component.

Potential derived features include:

Workload Intensity

$$\text{Workload Intensity} = \frac{\text{Patients per Shift}}{\text{Weekly Work Hours}}$$

Other derived features may include:

Work-hour category

Patient-load category

Occupational workload score

Department-level stress indicators

The exact transformations will be finalized after exploratory analysis.

2.4 DASS-42 Dataset

DASS-42 is a 42-item psychological questionnaire designed to measure three dimensions:

Depression

Anxiety

Stress

The 42 questions are divided into three 14-item subscales. Each question is generally scored on a four-point scale from 0 to 3.

Contribution

DASS-42 provides the psychological component. It can provide indicators for depression, anxiety and stress, and potentially relevant demographic/contextual information depending on the specific public dataset selected. Where such demographic attributes exist and are appropriate for analysis, they may also support the fairness evaluation described in Chapter 7.

2.5 Healthcare Workforce Mental Health Dataset (*New Addition*)

The fourth dataset is the Kaggle Healthcare Workforce Mental Health Dataset, which provides workplace stressor and workforce mental-health outcome information.

Potential variables (exact schema to be verified from the downloaded CSV)

Workplace stressors

Burnout-related indicators

Job satisfaction indicators

Absenteeism-related variables

Turnover-related variables

Role/department/experience context

Contribution

This dataset provides the workforce mental-health component, supporting:

Organization/department-level analysis

Additional features or auxiliary targets

Robustness and generalization experiments

The organization-oriented report view

Important dataset limitation

If the Kaggle source confirms that the dataset is synthetic, it must be clearly labeled as such. Results derived from it should be presented as experimental evidence about the *modeling framework*, not as clinical or population-level evidence about actual healthcare workers. It is an additional source of

information and an experimental validation resource, not a fourth set of rows that can automatically be aligned to the other three.

2.6 DASS-42 Target Categories

DASS-42 scores can be converted into severity categories.

Dimension	Normal	Mild	Moderate	Severe	Extremely Severe
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Depression	0–9	10–13	14–20	21–27	≥28
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Anxiety	0–7	8–9	10–14	15–19	≥20
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Stress	0–14	15–18	19–25	26–33	≥34
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The final target formulation will be selected after examining the actual class distribution in the chosen dataset. If severe class imbalance is observed, categories may be consolidated into fewer risk groups for a more reliable machine-learning experiment.

2.7 Dataset Integration Strategy

This is a critical part of the methodology.

The project will not directly concatenate the four datasets row-by-row. For example, we will not assume:

Stress-Lysis Row 1

+

Workplace Row 1

+

DASS-42 Row 1

+

Workforce Row 1

=

Same Healthcare Worker

because these datasets were collected independently.

Instead, the system will create a unified analytical representation:

Stress-Lysis

↓

Physical / Environmental Representation

Workplace Survey



Occupational Representation

DASS-42



Psychological Representation

Workforce Dataset



Workforce Representation



Unified Analytical Feature Framework

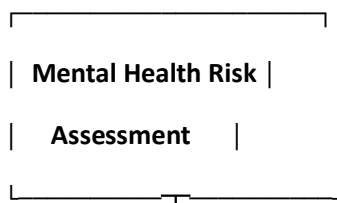
(harmonized schema + source metadata)

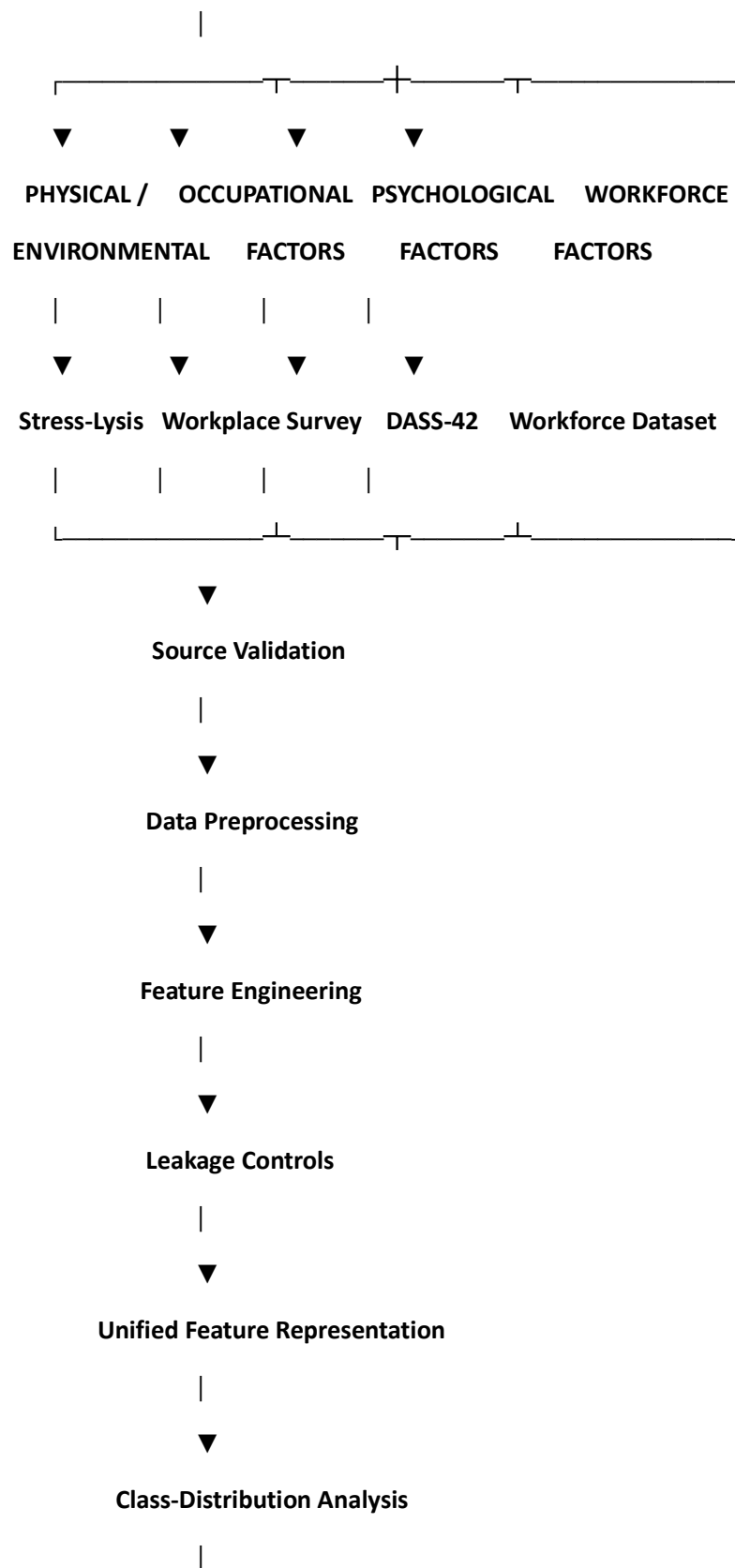
Each record retains a source/population label, which is later used for ablation studies, error analysis and dataset-shift discussion. This makes the methodology more scientifically defensible.

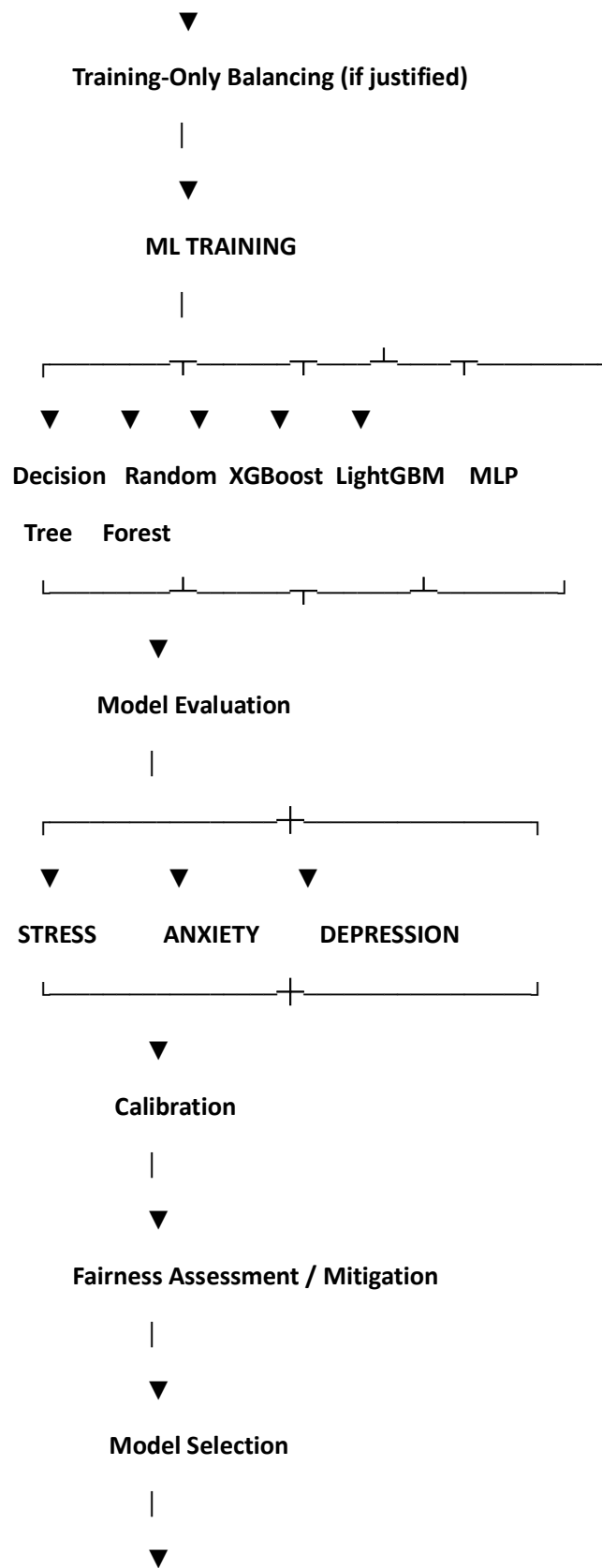
CHAPTER 3 — PROPOSED METHODOLOGY

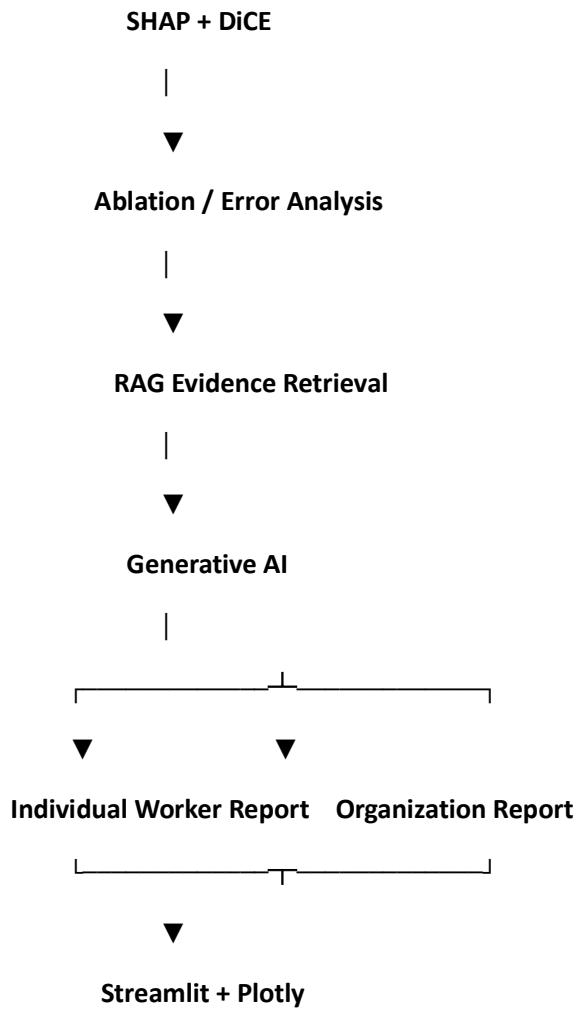
3.1 Overall Architecture

HEALTHCARE WORKERS









3.2 Data Preprocessing

The following steps will be performed:

Load datasets.

Inspect data types and validate schema.

Identify missing values and apply justified imputation.

Identify duplicate records.

Handle invalid values.

Encode categorical variables.

Calculate derived variables.

Analyze outliers.

Analyze class distribution per target.

Normalize/standardize numerical features where required.

Separate input variables and targets.

Perform a strict train/validation/test split before any learned preprocessing or resampling.

Retain dataset/source labels for later ablation and error analysis.

3.3 Feature Engineering

Feature engineering will convert raw variables into meaningful indicators.

Physical Features

Steps

Activity level / physical activity intensity

Environmental Features

Temperature

Humidity

Environmental stress indicator

Occupational Features

Weekly work hours

Patients per shift

Workload intensity

Work-hour category / patient-load category

Department and department-level contextual features

Psychological Features

DASS depression indicators

DASS anxiety indicators

DASS stress indicators

(subscale indicators constructed while preventing target leakage)

Workforce Features

Workplace stressor indicators

Burnout / job-satisfaction / absence / turnover-related variables, after exact CSV schema verification

Interaction Features

Only where supported by domain rationale and validated improvement

The exact final feature set will be determined through exploratory data analysis and validation.

3.4 Unified Feature Representation

Physical

+

Environmental

+

Occupational

+

Psychological

+

Workforce

↓

Harmonized Feature Schema

**(compatible scales, target definitions,
source/population metadata)**

↓

Machine Learning Models

The purpose is to allow the model to analyze multiple dimensions relevant to healthcare-worker mental health, while keeping each source identifiable so that its contribution can be tested experimentally.

3.5 Target Variables

The system will investigate three prediction tasks:

Stress Prediction — predict stress risk/severity.

Anxiety Prediction — predict anxiety risk/severity.

Depression Prediction — predict depression risk/severity.

These may be implemented as separate classification outputs or as coordinated outputs within the unified framework. Where confirmed in the fourth dataset, workforce indicators such as burnout or job satisfaction may serve as auxiliary analysis targets.

3.6 Data Leakage Prevention

Data leakage will be explicitly controlled.

For example, when predicting depression severity, the final depression score cannot be used as an input feature.

Instead:

DASS Questionnaire Responses

+

Relevant Contextual Features

↓

Depression Prediction

Similarly:

Target-derived features will not be allowed to leak into the input variables.

Scalers, encoders, imputers and resamplers will be fitted only on training data.

Resampling will never be applied to validation or test sets.

3.7 Class Imbalance

The class distribution of each target will be analyzed. If imbalance is significant, appropriate techniques may be investigated, including class weighting, stratified sampling, SMOTE where appropriate, and other resampling techniques. The technique will be selected based on the actual dataset distribution rather than being applied unnecessarily.

Every resampling method must be applied only to the training portion of the data. Validation and test sets must retain their original distributions.

Experiment	Purpose
Baseline	Measure model behavior without resampling
Class weighting	Change training loss/cost without synthesizing samples
SMOTE	Increase representation of minority classes in training data

Experiment	Purpose
Comparison	Determine whether balancing improves recall/F1 without unacceptable calibration or fairness degradation

CHAPTER 4 — MACHINE LEARNING MODELS

Five machine-learning algorithms will be compared.

4.1 Decision Tree

Decision Tree will be used as the basic interpretable baseline.

Advantages

Simple

Easy to visualize

Easy to explain

Handles nonlinear relationships

Disadvantage

Can overfit without parameter constraints.

Important parameters include: `max_depth`, `min_samples_split`, `min_samples_leaf`.

4.2 Random Forest

Random Forest combines multiple decision trees to improve robustness and generalization. It will be retained from the previous project to provide continuity.

Advantages include robustness, handling of nonlinear relationships, feature importance and good performance on structured data.

4.3 XGBoost

XGBoost is a gradient-boosting algorithm based on decision trees. It will be retained from the previous project because of its strong performance on structured/tabular datasets.

4.4 LightGBM

LightGBM is another gradient-boosting algorithm. It provides an important comparison with XGBoost.

Parameters that may be tuned include: `num_leaves`, `max_depth`, `learning_rate`, `n_estimators`, `min_child_samples`.

Care will be taken to control overfitting, especially because some of the datasets are relatively small.

4.5 Multilayer Perceptron

The existing MLP will be retained.

A possible architecture is:

Input

↓

Dense Layer – 64

↓

Dense Layer – 32

↓

Dense Layer – 16

↓

Output

Potential techniques: ReLU, Batch Normalization, Dropout, Adam optimizer, Early stopping.

The final architecture will be tuned according to the unified feature set.

4.6 Hyperparameter Tuning Strategy

Hyperparameter tuning will be performed using a controlled validation strategy, preferably stratified cross-validation for classification tasks. The exact search space will be reported so that the final comparison is reproducible, and tuning will be performed without touching the test set.

4.7 Model Comparison

Model	Type	Role
Decision Tree	Single tree	Interpretable baseline
Random Forest	Bagging ensemble	Robust tree ensemble
XGBoost	Gradient boosting	Existing high-performance model
LightGBM	Gradient boosting	New boosting comparison
MLP	Neural network	Nonlinear model comparison

5.1 Predictive Performance Metrics

Accuracy — measures overall classification correctness.

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

Precision — measures how many predicted positive cases are actually positive.

$$\text{Precision} = \frac{TP}{TP+FP}$$

Recall — measures how many actual positive cases are correctly identified.

$$\text{Recall} = \frac{TP}{TP+FN}$$

F1 Score — balances precision and recall.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Confusion Matrix — confusion matrices will be generated for each prediction task.

ROC-AUC — ROC-AUC will be used where the target formulation makes it appropriate.

5.2 Expanded Evaluation Framework

The evaluation is extended beyond predictive performance into four additional dimensions.

Evaluation dimension	Metrics / outputs	Question answered
Predictive performance	Accuracy, precision, recall, F1, ROC-AUC, confusion matrix	How well does the model classify?
Calibration	Brier score, ECE, reliability diagram	Do predicted probabilities correspond to observed frequencies?
Fairness	Group TPR/FPR differences, demographic parity measures, disparate-impact style ratios where appropriate	Does performance differ across groups?
Explainability	Global/local SHAP, DiCE counterfactuals	Why did the model produce this prediction?
Robustness	Dataset/source ablations, error analysis	Does the conclusion depend on a specific source or feature group?

5.3 Final Results Table

The final report will contain:

Model	Stress	Anxiety	Depression	Weighted F1	Brier	ECE	Fairness	Gap
Decision Tree	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Random Forest	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
XGBoost	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
LightGBM	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
MLP	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

Actual values will be inserted only after model training.

CHAPTER 6 — PROBABILITY CALIBRATION (*New Chapter*)

Calibration is added because a risk system should distinguish between a ranking model and a probability estimate. A model producing 0.80 risk should be tested for whether cases receiving approximately 0.80 probability actually exhibit the outcome at roughly that frequency in a suitable evaluation population.

6.1 Methods

Reliability diagram — compare predicted probability bins with observed event frequency.

Brier score — evaluate probabilistic prediction error.

Expected Calibration Error (ECE) — summarize the gap between predicted confidence and observed frequency across bins.

Platt/sigmoid scaling or isotonic regression — investigate post-hoc calibration using validation data only.

Report calibration before and after calibration rather than assuming calibration automatically improves.

6.2 Evaluation Protocol

Calibration will be evaluated separately for each target, because stress, anxiety and depression may have different class distributions and model behavior.

Calibration will be fitted on a validation split and reported on the held-out test split.

Calibration will be re-measured after class balancing, since resampling can distort predicted probabilities.

Instability of calibration estimates on small test sets will be explicitly acknowledged.

6.3 Output

Calibrated probabilities feed directly into:

The individual risk report

The low-confidence / uncertainty flag

The Model Comparison dashboard page

CHAPTER 7 — FAIRNESS AND BIAS MITIGATION (*New Chapter*)

The original proposal identified bias as an ethical consideration and stated that potential dataset biases should be acknowledged. The expanded framework operationalizes this requirement with explicit measurement and mitigation using Fairlearn.

7.1 Fairness Workflow

Identify demographic/contextual attributes actually available and appropriate for analysis in the chosen datasets.

Evaluate whether sample sizes are sufficient for meaningful group comparisons.

Measure group-specific recall, false-positive rate, false-negative rate and related fairness statistics.

Use Fairlearn to visualize and compare performance across groups.

Where justified, investigate mitigation such as reweighting, threshold optimization or other constrained approaches.

Compare baseline versus mitigated models across performance, calibration and fairness together, rather than optimizing one metric in isolation.

7.2 Constraints on the Analysis

Fairness analysis must not invent demographic variables that are absent from the source data. If a dataset does not contain an appropriate sensitive attribute, the report will state clearly that fairness for that dimension could not be evaluated.

Where group sample sizes are too small for stable estimates, the report will present the result as indicative only.

7.3 Reported Outputs

Group-wise performance table

Fairness gap metrics (e.g. equalized-odds difference, demographic parity difference)

Baseline vs mitigated comparison

CHAPTER 8 — EXPLAINABLE AI

8.1 SHAP

SHAP will be used to understand how different features influence model predictions.

The goal is not only:

"What did the model predict?"

but also:

"Why did the model make this prediction for this healthcare worker?"

8.2 Global Explainability

Global SHAP analysis will identify the most important features across the dataset.

Potential categories include:

Occupational factors

Psychological factors

Physical factors

Environmental factors

Workforce factors

The actual feature rankings will be obtained experimentally.

8.3 Local Explainability

For an individual healthcare worker:

Predicted Stress Risk: High

SHAP can provide:

Feature A → increased prediction

Feature B → increased prediction

Feature C → decreased prediction

This allows the system to provide an explanation rather than simply returning a classification.

8.4 Explainability Across Models

SHAP-based explanations will be investigated for Decision Tree, Random Forest, XGBoost, LightGBM and MLP. The appropriate SHAP explainer (TreeExplainer, KernelExplainer, DeepExplainer, etc.) will be selected based on the model type.

8.5 DiCE Counterfactual Explanations (*New Addition*)

DiCE adds a complementary explanation question. Instead of only asking *why* the model predicted a given risk, the system can investigate *what feasible input changes would alter the model's prediction*.

Generate counterfactual examples for suitable tabular models.

Apply feasibility constraints so unrealistic changes are not suggested.

Separate actionable variables (e.g. weekly work hours, activity level) from immutable/contextual variables (e.g. department, demographic attributes).

Present counterfactuals as model-based scenarios, not medical prescriptions.

Example format

Current prediction: Stress Risk = High

Counterfactual scenario:

Weekly work hours: 64 → 52

Patients per shift: 18 → 14

→ Predicted Stress Risk = Moderate

(Model-based what-if scenario only.

Not a causal claim or a clinical recommendation.)

8.6 Explanation Handoff Rule

Only model-generated evidence — prediction, calibrated probability, SHAP contributions and DiCE counterfactuals — is passed into the reporting layer. The reporting layer does not compute or modify risk.

CHAPTER 9 — ABLATION STUDY DESIGN (*New Chapter*)

Ablation studies provide direct evidence for whether each data source and methodological component contributes to the final framework.

Ablation group	Example configurations
Dataset ablation	Stress-Lysis only; Workplace only; DASS only; fourth dataset only; selected combinations; all sources
Feature-group ablation	Physical/environmental; occupational; psychological; workforce; all groups
Balancing ablation	No balancing vs class weighting vs SMOTE
Calibration ablation	Uncalibrated vs calibrated
Fairness ablation	Baseline vs mitigation
Explanation ablation	SHAP-only vs SHAP + DiCE
Report ablation	Direct GenAI vs grounded RAG + model evidence

For every ablation, the same evaluation metrics are recorded so that improvements and trade-offs can be discussed quantitatively. Negative results will be reported rather than hidden.

CHAPTER 10 — GENERATIVE AI AND RAG

The existing Generative AI component will be retained and expanded with a Retrieval-Augmented Generation (RAG) layer, so that narrative recommendations are grounded in a controlled evidence collection.

10.1 Inputs to the Generative Layer

The ML system produces:

Prediction

+

Calibrated Probability

+

Uncertainty / Confidence Flag

+

SHAP Feature Contributions

+

DiCE Counterfactual Scenarios

+

Structured Context (department, workload, etc.)

10.2 Proposed RAG Flow

Model prediction + calibrated probability + SHAP + DiCE + structured context



Retrieval of approved evidence passages



Prompt = model evidence | retrieved evidence



Generative AI



Report with source-aware language

10.3 RAG Requirements

Create a small curated evidence corpus from appropriate public/academic/organizational sources on occupational stress and workplace wellbeing.

Retrieve relevant passages based on the report topic.

Give the LLM the prediction and explanations separately from retrieved evidence.

Instruct the LLM not to change the model prediction.

Require cautious language for uncertainty and limitations.

Clearly distinguish model-derived observations from general evidence and recommendations.

Include a non-diagnostic disclaimer in every generated report.

10.4 Example

Input

Healthcare Worker

Department: ICU

High workload

Long working hours

Elevated psychological indicators

Model Output

Stress Risk: High (p = 0.83, calibrated)

Anxiety Risk: Moderate (p = 0.54, calibrated)

Depression Risk: Low (p = 0.18, calibrated)

SHAP

Workload → increases stress prediction

Work Hours → increases stress prediction

Feature X → increases anxiety prediction

Feature Y → decreases risk

DiCE

Reducing weekly hours and patients per shift

moves the predicted stress category downward.

Generative AI

The system converts these results into an understandable, evidence-grounded summary without altering the prediction.

10.5 Safety

The system must never claim to diagnose a healthcare worker. It should state:

"The prediction represents a machine-learning-based risk estimate and is not a clinical diagnosis."

The system is intended for research, education, awareness and experimental prediction.

RAG does not make a report medically authoritative — the quality of retrieved evidence remains a limitation, and the Generative AI layer may hallucinate unless tightly constrained and evaluated.

CHAPTER 11 — TWO REPORT TYPES (*New Chapter*)

11.1 Individual / Healthcare Worker Report

Stress, anxiety and depression risk estimates.

Calibrated probabilities where supported by evaluation.

Risk category and uncertainty/low-confidence flag where appropriate.

Top SHAP contributing factors.

Counterfactual scenarios where feasible.

General, non-clinical wellness information grounded through RAG.

Clear disclaimer that the result is a model-based risk estimate, not a diagnosis.

11.2 Organization / Workforce Report

Aggregate department-level patterns where sample sizes support reporting.

Stress/burnout and other workforce indicators available in the fourth dataset.

Workload and job-satisfaction patterns where available.

Model performance and calibration summary.

Fairness comparison across supported groups.

Top global SHAP factors.

Error and uncertainty analysis.

Evidence-grounded organizational considerations rather than individualized medical recommendations.

Aggregate reporting will be used wherever individual privacy or small sample sizes make individual reporting inappropriate.

CHAPTER 12 — STREAMLIT + PLOTLY APPLICATION

The existing Streamlit application will be expanded rather than rebuilding the entire application architecture. There is no requirement for a separate frontend, backend or database for this project. Plotly is added *inside* the same application for interactive analytical visualization.

12.1 Proposed Interface

AURA WELLNESS

|

| — Home

| — Purpose, workflow, limitations

|

| — Individual Assessment

| — Stress Prediction

	— Anxiety Prediction
	— Depression Prediction
	— Explainability
	— SHAP (global / local)
	— DiCE Counterfactuals
	— Model Comparison
	— Performance
	— Calibration
	— Fairness
	— Dataset / Ablation Analysis
	— Organization Analytics
	— AI Generated Report
	— Individual Report
	— Organization Report
	— Responsible AI
	— Fairness, calibration, limitations, data-source caveats

12.2 Page Contents

Page / module	Main contents
Home	Project purpose, limitations and workflow
Individual Assessment	Inputs, stress/anxiety/depression predictions, probabilities and explanations

Page / module	Main contents
Explainability	SHAP plots and DiCE counterfactuals
Model Comparison	Performance, calibration and fairness charts
Dataset / Ablation Analysis	Feature-group and dataset experiment comparisons
Organization Analytics	Department/workforce-level aggregates and patterns
AI Report	RAG-grounded individual or organization report
Responsible AI	Fairness, calibration, limitations and data-source caveats

12.3 User Workflow

Healthcare Worker Data



Data Preprocessing



Feature Engineering



Unified Feature Representation



Five ML Models



Mental Health Predictions



Calibrated Probabilities



SHAP + DiCE Explanation



RAG Evidence Retrieval



Generative AI



Individual / Organization Report



Plotly Interactive Analytics

CHAPTER 13 — SYSTEM REQUIREMENTS

13.1 Software / Technology Stack

Layer	Technology / method	Purpose
Programming	Python	Core implementation
Environment	Jupyter Notebook	Experimentation
Data	Pandas, NumPy	Loading, cleaning and feature engineering
ML	scikit-learn	Preprocessing, evaluation and classical models
Boosting	XGBoost, LightGBM	Gradient-boosted models
Neural network	TensorFlow/Keras	MLP
Imbalance	SMOTE / class weighting	Minority-class handling where justified
XAI	SHAP	Global/local feature attribution
Counterfactual XAI	DiCE	What-if explanations
Fairness	Fairlearn	Bias measurement and mitigation experiments
Calibration	scikit-learn / custom ECE	Probability reliability
Experiment evaluation	Ablation framework	Quantify component contributions
Visualization	Matplotlib, Seaborn, Plotly	Static and interactive charts
Application	Streamlit	Interactive interface
GenAI	Generative AI API	Natural-language reports
Knowledge grounding	RAG	Evidence-grounded report generation

Layer	Technology / method	Purpose
Experiment tracking	MLflow (<i>optional</i>)	Tracking of runs, parameters and metrics

13.2 Hardware

The project does not require high-end infrastructure.

Recommended:

8 GB RAM minimum

16 GB RAM preferred

Modern multi-core CPU

GPU optional

13.3 Optional: MLflow

MLflow is not essential to the scientific core, but it is a useful engineering layer if time permits. Each experiment can record the dataset configuration, feature group, balancing method, model, hyperparameters, performance metrics, calibration metrics and fairness metrics.

A useful experiment naming structure is:

`target_datasetConfiguration_balance_model_calibration_fairness`

This makes it easier to demonstrate that the final result came from a controlled experiment rather than a single manually selected run.

CHAPTER 14 — IMPLEMENTATION PLAN

Phase	Activities
Phase 1 — Baseline Reproduction	Clone existing project; verify existing notebooks; reproduce existing results; verify Streamlit application.
Phase 2 — Dataset Acquisition	Obtain DASS-42 and the fourth workforce dataset; understand questionnaire structure; calculate/verify DASS scores; verify schemas and licenses.
Phase 3 — Data Audit	Missing values, duplicates, target distributions, source/population metadata, leakage risks, synthetic-data status.
Phase 4 — Dataset Processing	Process Stress-Lysis, workplace data, DASS-42 and the workforce dataset; standardize features; handle missing values.
Phase 5 —	Build source-specific representations and a controlled unified analytical

Phase	Activities
Harmonization	schema; establish prediction targets; prevent data leakage.
Phase 6 — Feature Engineering	Create justified physical, environmental, occupational, psychological and workforce features.
Phase 7 — Baseline ML	Train Decision Tree, Random Forest, XGBoost, LightGBM and MLP using controlled train/validation/test procedures.
Phase 8 — Imbalance Experiments	Compare baseline, class weighting and SMOTE where needed (training data only).
Phase 9 — Evaluation	Accuracy, precision, recall, F1-score, confusion matrix, ROC-AUC where applicable.
Phase 10 — Calibration	Generate reliability curves; calculate Brier/ECE; test Platt/isotonic calibration; report before and after.
Phase 11 — Fairness	Measure group disparities with Fairlearn and test appropriate mitigation.
Phase 12 — Explainability	Implement global SHAP, local SHAP, feature importance and constrained DiCE counterfactuals.
Phase 13 — Ablation	Run dataset, feature, imbalance, calibration, fairness and explanation ablations.
Phase 14 — RAG + GenAI	Build curated evidence corpus, retrieval pipeline and grounded report prompts; integrate prediction, probability, SHAP, DiCE and AI summary.
Phase 15 — Dashboard	Extend Streamlit with Plotly visualizations and individual/organization views.
Phase 16 — Final Evaluation	Compare all five models across predictive, calibration, fairness and explainability outcomes; analyze errors, important features, strengths and limitations.
Phase 17 — Documentation	Report limitations, synthetic-data caveats, ethical considerations and reproducibility.

CHAPTER 15 — EXPECTED RESULTS

The project is expected to produce:

A four-source unified analytical framework for healthcare-worker mental-health risk prediction.

Prediction across three dimensions: Stress, Anxiety, Depression.

A five-model predictive comparison.

Quantitative evidence from dataset and feature ablation studies.

A class-imbalance comparison using justified balancing methods.

Calibration analysis showing whether risk probabilities are reliable.

Fairness analysis identifying supported group-level disparities and mitigation effects.

SHAP-based global and local explanations.

DiCE counterfactual explanations where feasible.

Two report types for individual and organization-oriented interpretation.

RAG-grounded Generative AI reports that do not independently determine model predictions.

An interactive Plotly/Streamlit application.

A clear statement of the limits of synthetic, heterogeneous and non-clinically validated data.

CHAPTER 16 — EXPECTED CONTRIBUTION

The main contribution is the transition from a stress and burnout prediction system for healthcare workers into a broader, responsible and experimentally validated mental-health risk prediction framework for healthcare workers.

The framework considers four perspectives:

PHYSICAL / ENVIRONMENTAL



Stress-Lysis

OCCUPATIONAL



Healthcare Workplace Survey

PSYCHOLOGICAL



DASS-42

WORKFORCE



Healthcare Workforce Mental Health Dataset

These complementary sources are transformed into a unified analytical framework.

The project additionally compares five different machine-learning algorithms and incorporates class-imbalance experiments, calibration, fairness assessment, Explainable AI (SHAP + DiCE), ablation studies, RAG-grounded Generative AI and interactive analytics.

CHAPTER 17 — LIMITATIONS

17.1 Different Data Sources

The four datasets originate from different sources and populations. Therefore, they should not be interpreted as one naturally collected healthcare-worker cohort, and they do not represent a longitudinal cohort.

17.2 Different Feature Spaces

The datasets measure different aspects of human behavior and mental health. Therefore, feature engineering and normalization are required before constructing the unified framework.

17.3 DASS-42 Dataset Population

DASS-42 itself is a psychological assessment instrument and is not inherently restricted to healthcare workers. Therefore, the report should not falsely claim that every DASS-42 respondent is a healthcare worker. Instead, DASS-42 serves as the psychological component of the framework, while the project's application domain remains healthcare workers.

17.4 Synthetic Fourth Dataset

The Healthcare Workforce Mental Health dataset may be synthetic. If so, its results must not be interpreted as real-world prevalence or clinical evidence, and its role is restricted to methodological and workforce-analysis experiments.

17.5 Dataset Shift

Different source distributions can create dataset shift, which may affect generalization across sources.

17.6 Self-Reported Information

DASS-42 is based on self-reported responses, which may introduce reporting bias.

17.7 Fairness Limitations

Fairness analysis is limited by the sensitive attributes actually present in the data and by the sample sizes available for each group.

17.8 Calibration Limitations

Calibration estimates may be unstable for small test sets, and calibration fitted on one source may not transfer to another.

17.9 Resampling Limitations

SMOTE can create synthetic minority samples that do not necessarily correspond to real individuals.

17.10 Counterfactual Limitations

Counterfactuals are model-based what-if scenarios, not guaranteed causal effects.

17.11 Generative AI Limitations

RAG does not make a report medically authoritative; retrieved evidence quality remains important. Generative AI may hallucinate unless tightly constrained and evaluated.

17.12 No Clinical Diagnosis

The system is not clinically validated and should not be used to diagnose mental-health disorders.

CHAPTER 18 — ETHICAL AND RESPONSIBLE-AI CONSIDERATIONS

Privacy — Personal and identifiable information should not be unnecessarily collected or exposed.

Responsible Prediction — Predictions must be presented as risk estimates rather than diagnoses.

Transparency — The system should explain the factors influencing predictions, and show uncertainty and limitations where appropriate.

Bias — Potential biases within datasets should be measured where possible and acknowledged where measurement is not possible. Fairness mitigation must not be used to conceal poor data quality.

Generative AI — The Generative AI system should explain model outputs rather than make medical claims, and must not override model outputs. Model predictions, explanations and retrieved evidence must remain distinguishable in the report.

Aggregation — Use aggregate organization analytics where individual privacy or small sample sizes make individual reporting inappropriate.

Documentation — Synthetic-data usage must be documented explicitly.

Honest Reporting — Negative results and trade-offs must be reported rather than selecting only favorable experiments.

No Guaranteed Outcomes — Do not generate recommendations that imply guaranteed clinical outcomes.

CHAPTER 19 — FUTURE SCOPE

The project could later be extended with:

Wearable Data

Heart rate, HRV, sleep, activity, physiological signals

Longitudinal Monitoring

Day 1 → Day 2 → Day 3 → Day 4 → Mental Health Trend

Real-Time Healthcare Worker Monitoring

Future systems could potentially monitor changes in risk over time.

Mobile Application

The Streamlit prototype could later be converted into a mobile or web application.

Advanced Deep Learning

Future models could include LSTM, GRU and Transformer architectures.

Multimodal Mental Health

Questionnaires + Wearables + Behavioral Data + Occupational Data + Text

Causal Analysis

Moving from counterfactual what-if scenarios toward causal inference methods.

CHAPTER 20 — PROJECT NOVELTY

The project's novelty is:

A unified, explainable, fair and calibrated multi-source machine-learning framework specifically focused on mental-health risk prediction among healthcare workers, incorporating physical, environmental, occupational, psychological and workforce information, and grounding its generated reports in retrieved evidence.

The novelty does not come from inventing new ML algorithms. Instead, it comes from the integration and experimental evaluation of several complementary layers.

Novelty component	Contribution
Multi-source	Four heterogeneous sources representing different information dimensions
Multi-task	Stress, anxiety and depression risk assessment
Responsible AI	Explicit fairness and calibration analysis
Explainable AI	SHAP plus counterfactual DiCE explanations
Experimental rigor	Ablation studies and controlled comparisons
Evidence-grounded GenAI RAG separates evidence retrieval from model prediction	
Stakeholder views	Individual and organization-oriented reports
Interactive analytics	Plotly + Streamlit model and workforce analysis
Continuity	Direct extension of the previous semester's system

CHAPTER 21 — FINAL SCOPE JUSTIFICATION

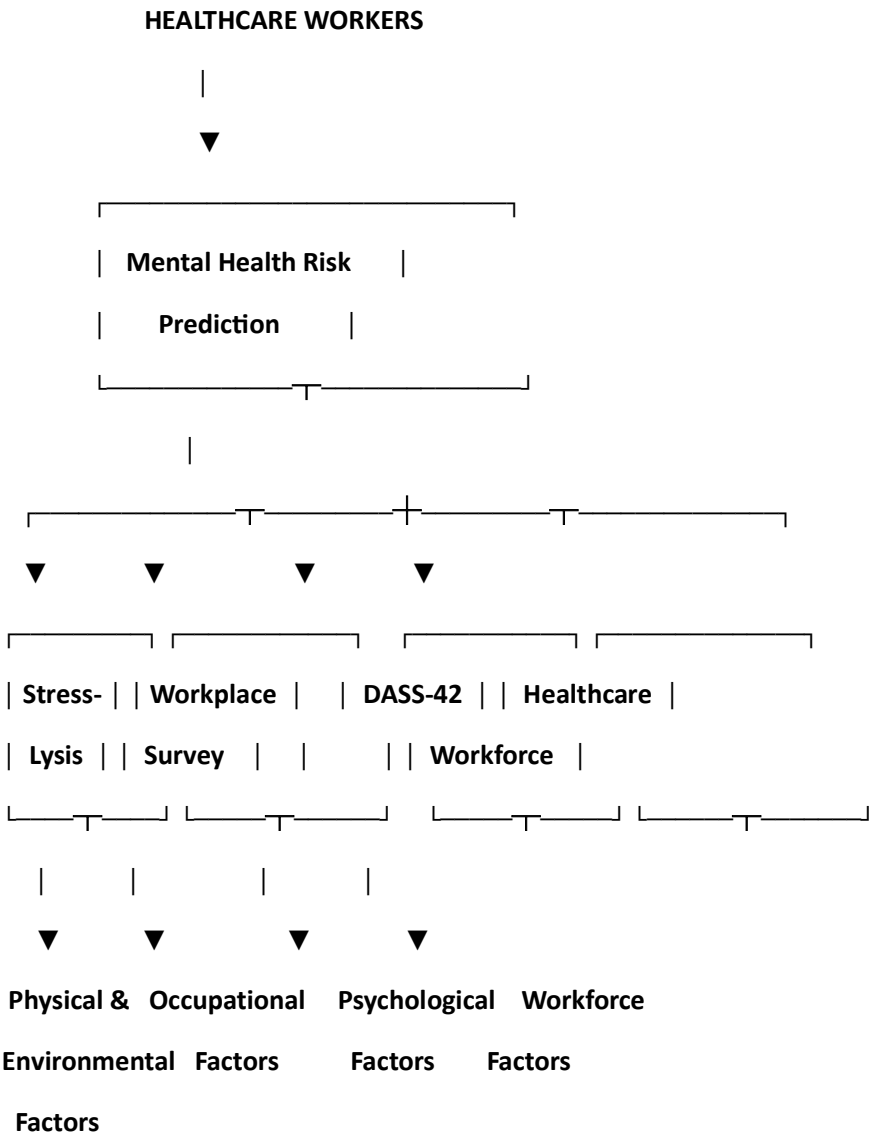
The strongest version of the project is not the one with the greatest number of technologies. It is the one in which every technology answers a research question. The proposed additions have clear roles:

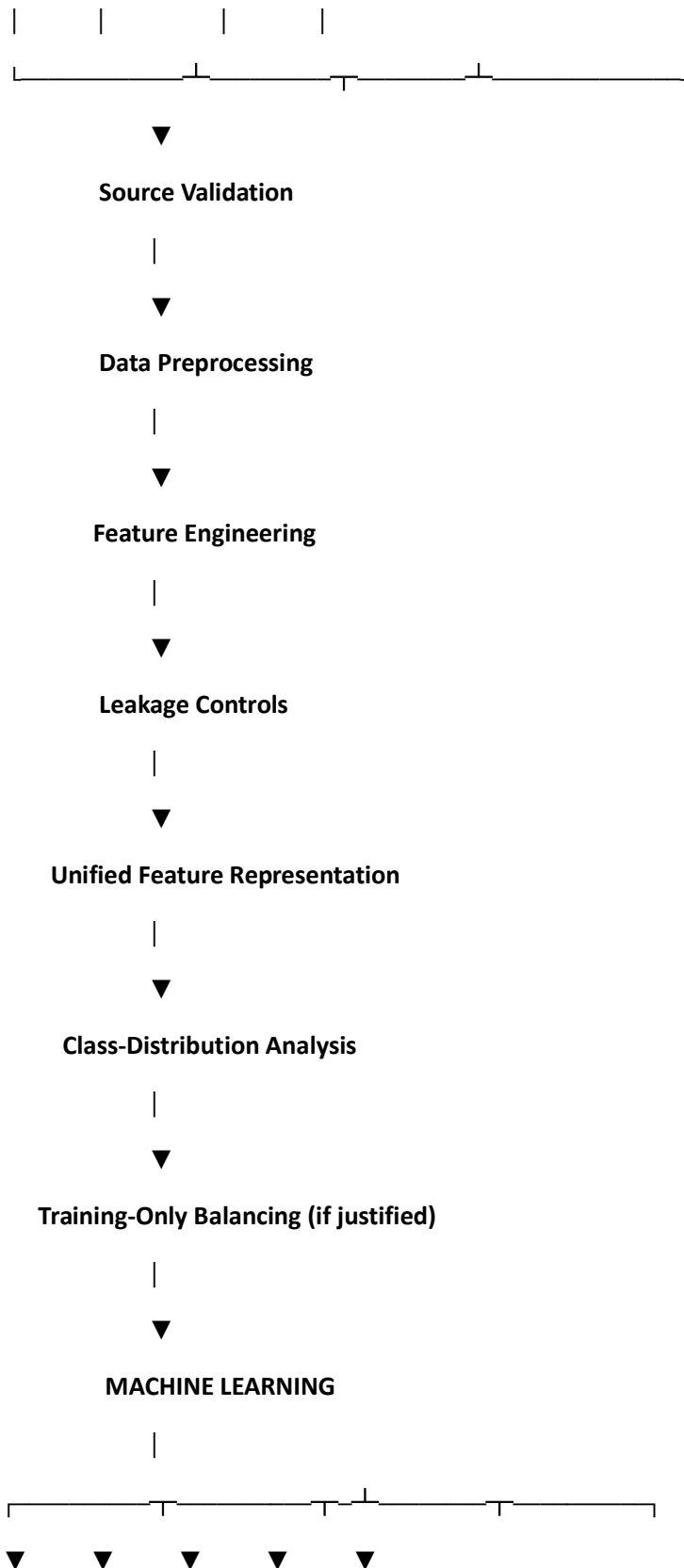
Component	Research question it supports
Fourth dataset	Does an additional workforce-oriented source improve robustness or add complementary information?
SMOTE / class weighting	How does imbalance handling affect minority-class detection?
Fairlearn	Are predictions and errors distributed differently across supported groups, and can mitigation help?
Calibration	Can the model probabilities be trusted as probabilities?
SHAP	Which features drive predictions?
DiCE	What feasible input changes alter model predictions?
Ablation	Which data sources and methodological components actually matter?
RAG	Can generated reports be grounded in retrieved evidence rather than free-form

Component	Research question it supports
	generation?
Plotly	Can complex model, fairness and workforce results be explored interactively?
MLflow (optional)	Can the final result be shown to come from a controlled, reproducible experiment?

This structure turns the project into a coherent experimental framework rather than a collection of unrelated features, while preserving the original direction, architecture and non-clinical positioning.

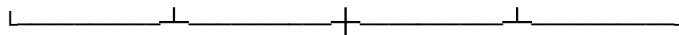
FINAL SYSTEM ARCHITECTURE



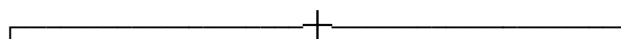


Decision Random XGBoost LightGBM MLP

Tree Forest



Model Evaluation



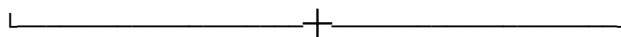
STRESS



ANXIETY



DEPRESSION



Calibration

(Brier / ECE / Reliability)



Fairness Assessment & Mitigation

(Fairlearn)



Model Selection

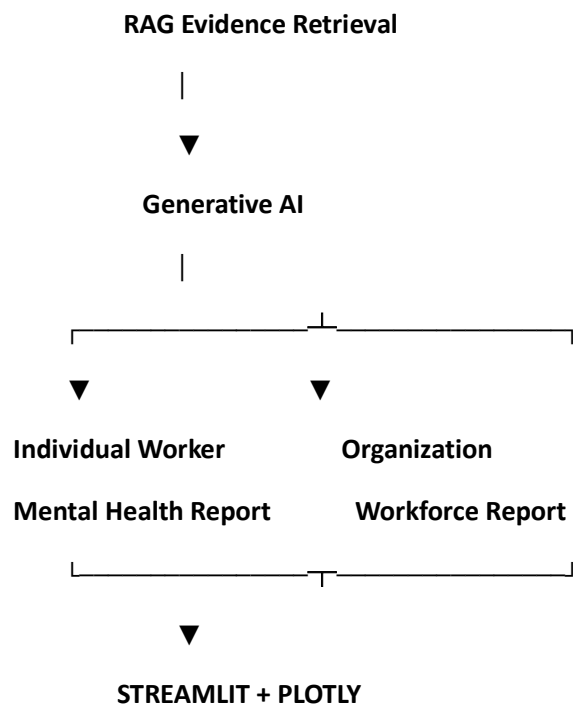


SHAP + DiCE



Ablation & Error Analysis





FINAL END-TO-END WORKFLOW

Healthcare-worker data / research datasets

- Data audit
- Source-specific preprocessing
- Feature engineering
- Leakage control
- Unified analytical representation
- Class-distribution analysis
- Training-only SMOTE / class weighting when justified
- DT / RF / XGBoost / LightGBM / MLP
- Performance evaluation
- Calibration
- Fairness analysis & mitigation
- Model comparison

- SHAP
 - DiCE
 - Ablation & error analysis
 - Calibrated risk outputs
 - RAG retrieval
 - Evidence-grounded Generative AI
 - Individual / organization report
 - Plotly visual analytics
 - Streamlit application
-

FINAL PROJECT EVOLUTION

Previous Semester

Explainable Generative AI for Stress and Burnout Risk Prediction in Healthcare Workers

Healthcare Workers



2 Datasets



Stress + Burnout



Random Forest, XGBoost, MLP



SHAP



Generative AI



Streamlit

Current Semester

Mental Health Prediction Using Machine Learning for Healthcare Workers

Healthcare Workers



4 Datasets



Stress-Lysis + Healthcare Workplace Survey

+ DASS-42 + Healthcare Workforce Dataset



Unified Multi-Source Framework



Stress + Anxiety + Depression



Decision Tree, Random Forest, XGBoost, LightGBM, MLP



Class Imbalance Experiments (SMOTE / class weighting)



Calibration (Brier / ECE / reliability)



Fairness (Fairlearn)



SHAP + DiCE



Ablation Studies



RAG-Grounded Generative AI



Individual + Organization Reports



Streamlit + Plotly (MLflow optional)

Comparison Table

Previous / Existing Direction	Current Expanded Direction
2 datasets	4 complementary datasets
Stress + burnout focus	Stress + anxiety + depression, with workforce-level indicators
RF, XGBoost, MLP	Decision Tree, RF, XGBoost, LightGBM, MLP
Standard ML metrics	Performance + calibration + fairness + error analysis
SHAP explanations	SHAP + DiCE counterfactual explanations
Single AI report	Individual and organization-oriented reports
Generative AI	RAG + evidence-grounded Generative AI
Streamlit	Streamlit + Plotly interactive analytics
No explicit fairness layer	Fairlearn-based bias assessment and mitigation
No systematic component study	Ablation and controlled comparison experiments
No experiment tracking	Optional MLflow tracking layer

The expansion can therefore be summarized in one line:

The project evolves from predicting stress and burnout among healthcare workers to a broader, explainable, fair and calibrated machine-learning framework for predicting stress, anxiety and depression risk among healthcare workers using complementary physical, occupational, psychological and workforce information, with evidence-grounded report generation.
